



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Final-Term Exam: Trimester: Spring 2025 (I Grade)

Course Code: EEE 2123; Course Title: Electronics

Total Marks: 40; Duration: 2 hours

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

There are six (6) questions in this paper. Answer all of them.

Q1(a). The threshold voltage of each MOSFET in the following figure 1 is $V_t = 0.5$ V. Determine the region of operation of the MOSFET in each circuit. [4]

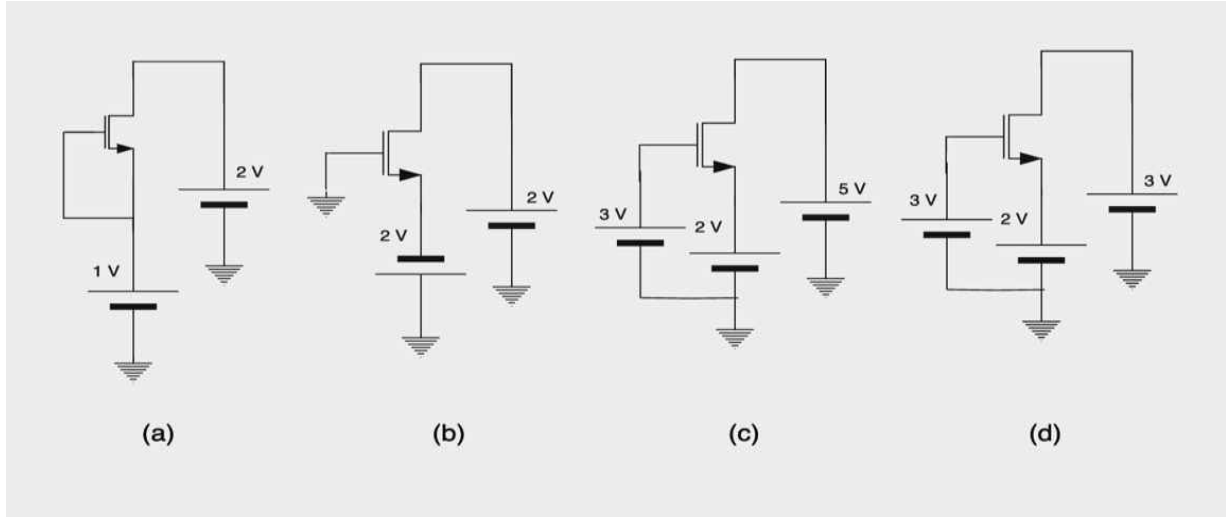


Fig 1: Circuit diagram for Q1(a)

Q1(b). For the circuit in Fig 2, find the largest value of R_D for which the transistor remains in saturation mode. The MOSFET has $V_t = 1$ V, $\mu_n C_{ox} = 0.5$ mA/V², $W/L = 2$. [6]

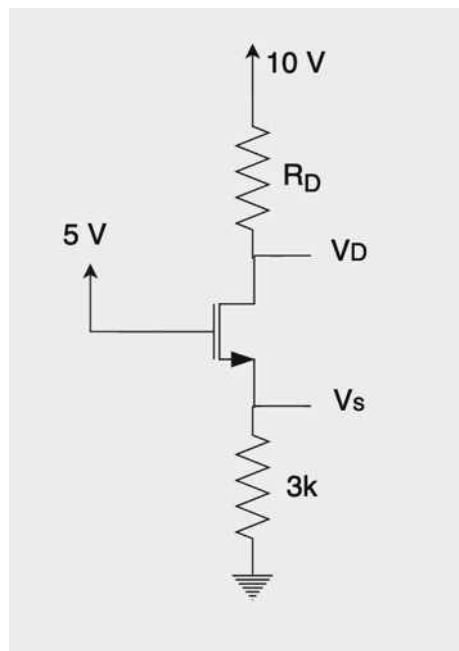


Fig 2: Circuit diagram for Q1(b)

Q2. Obtain the value of V_D , V_G and oxide capacitance from the circuit in Fig. 3 where effective resistance between drain and source is $500\ \Omega$ and $I_D = 0.4\text{ mA}$. The NMOS transistor has $V_t = 1\text{ V}$, $k'_n = 0.5\text{ mA/V}^2$, $W/L = 2$, $\mu_n = 450\text{ cm}^2/\text{Vs}$ [5]

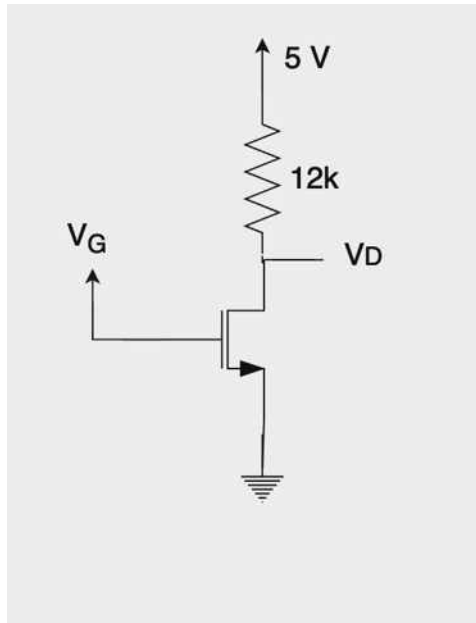
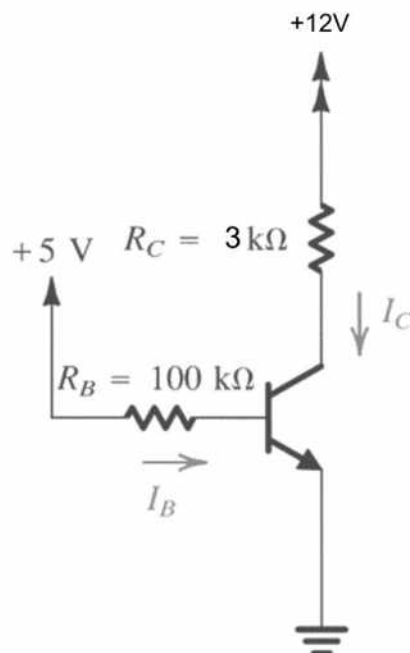


Fig 3: Circuit diagram for Q2

Q3. Find all node voltages and branch currents in the following circuit, $\beta = 155$ [8]



Q4. Find all node voltages and branch currents in the following circuit in Fig. 5.
 $\beta=100$ and $\beta_{\text{forced}} = 25$.

Assume,
[7]

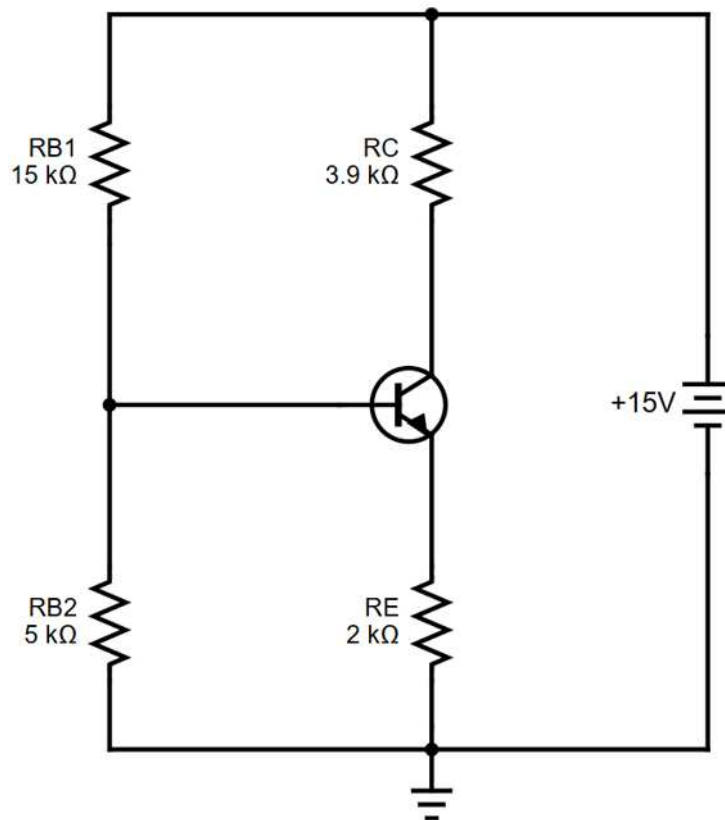


Fig 5: Circuit diagram for Q4

Q5(a). Implement the following logic in CMOS technology using a minimum number of transistors.
 Also write the total number of transistors used in your circuit. **[5]**

$$f = \underline{ABC + BD + AD + BC + A\underline{B}C}$$

Q5(b).

- (i) Write briefly why CMOS technology is most preferred in terms of power consumption. [1]
(ii) For the following PMOS network in Fig 6, draw the corresponding NMOS network for a CMOS Circuit. Also determine the output Y. [4]

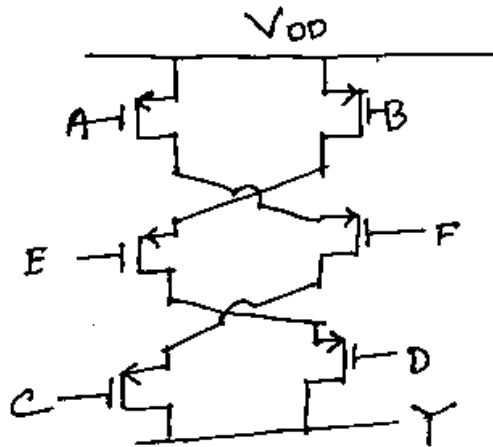


Fig 6: Circuit diagram for Q5(b)

Q6. Implement the equations given below using Op-Amps. Clearly mention all the values of resistances and capacitances.

(a) $V_o = 7 \int V_i dt + 10V_i - 25 \frac{d^2 V_i}{dt^2}$

In (a), V_i is the input signal. You **cannot use more than 5 Op-Amps** to implement (a). [5]

(b) $V_o = \sin(t) + 2\cos(t)$

In (b), The only input to this circuit is $\sin(t)$. [Hint: $\cos(t)$ can be produced from $\sin(t)$ by differentiation] [5]